Домашняя работа №1

№14.10

Дано:
$\bar{R}$, $\bar{M_0} = 0$
необх. и дост. услов. равн

Решение:

$\bar{R} = 0$, $\bar{M_0} = 0$

Надо доказать, что $\delta A_{акт} = \sum_i F_i \delta r_i = 0$

$\delta r_i = \delta r_{уm} + \delta \bar{\alpha} \cdot r_i$

r_i – радиус-вектор нкот. точки (к котор. прилож. F_i), δr_i – вирт. перемещ.

$\sum F_i \delta r_i = \sum_i F_i (\delta r_{уm} + \delta\alpha \cdot r_i) = \sum_i F_i \delta r_{уm} +$
$+ \sum_i F_i \delta\alpha \cdot r_i = \delta r_{уm} \cdot \sum_i F_i + \delta\alpha \sum_i F_i \cdot r_i =$

$(*) = \delta r_{уm} \cdot R + \delta\alpha \cdot M_0 = 0 + 0 = 0 \Rightarrow \delta A_{акт} = 0 \Rightarrow$

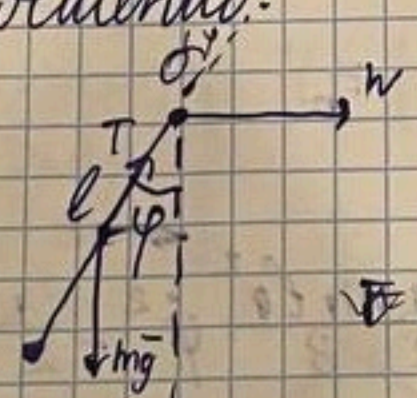
$\Rightarrow \bar{R} = 0, \bar{M_0} = 0$ достат. условие ПР.

Также это и необходим. услов., т.к δr_i может быть любым $\Rightarrow$ вариант, что сумма $(*)$ равна нулю, а по отдельности $R \neq 0$, $M_0 \neq 0$ невозможен. ч.т.д

№14.20

Дано:
w
ПР – ?

Решение:



$T_{абс} = T_{отн} + \frac{m\dot\varphi^2}{2} = \frac{ml^2}{3}\dot\varphi^2 +$

$r_1 = \left(\frac{l}{2}\sin\varphi, \frac{l}{2}\cos\varphi\right)$, $r_2 = \left(\frac{l}{2}\sin\varphi, \frac{l}{2}\cos\varphi\right)$

$F_1 = (0, mg)$, $\frac{\partial r_1}{\partial\varphi} = \left(\frac{l}{2}\cos\varphi, -\frac{l}{2}\sin\varphi\right) = \frac{\partial r_2}{\partial\varphi}$

$F_2 = (mw, 0)$

$Q_\varphi = \left(F_1, \frac{\partial r_1}{\partial\varphi}\right) + \left(F_2, \frac{\partial r_2}{\partial\varphi}\right) = -mg\frac{l}{2}\sin\varphi + mw\frac{l}{2}\cos\varphi = 0$

$g\sin\varphi = w\cos\varphi$

$tg\varphi = \frac{w}{g} \Rightarrow y = x \cdot ctg\varphi$, $y = x\frac{g}{w}$

Ответ: $y = x\frac{g}{w}$

Дано:

$f(x,y,z)=0$

$\{F_x(x,y,z), F_y, F_z\}$

ПР

№14.29

Решение:

$F_x + \lambda\frac{\partial f}{\partial x} = 0,\quad F_y + \lambda\frac{\partial f}{\partial y} = 0,\quad F_z + \lambda\frac{\partial f}{\partial z} = 0$

ПР $\Leftrightarrow F_i = 0 \Rightarrow \frac{\partial \Pi}{\partial x} = \frac{\partial \Pi}{\partial y} = \frac{\partial \Pi}{\partial z} = 0$

$L = \Pi + \lambda f$ — функция Лагранжа, ищем услов. экстремум при условии связи $f(x,y,z)=0$

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial y} = \frac{\partial L}{\partial z} = 0$$

$$\frac{\partial \Pi}{\partial x} + \lambda\frac{\partial f}{\partial x} = \frac{\partial \Pi}{\partial y} + \lambda\frac{\partial f}{\partial y} = \frac{\partial \Pi}{\partial z} + \lambda\frac{\partial f}{\partial z} = 0$$

$$F_x + \lambda\frac{\partial f}{\partial x} = 0,\quad F_y + \lambda\frac{\partial f}{\partial y} = 0,\quad \frac{\partial F}{\partial z} + \lambda\frac{\partial f}{\partial z} = 0$$

Точка является усл. экстр., если она — стационарная (фун. Лагранжа). ПР соответ. точки усл. экстремума $\Rightarrow$ решение системы определяет ПР и любому ПР соответ. решен. этой системы.

№ 14.37

Дано:

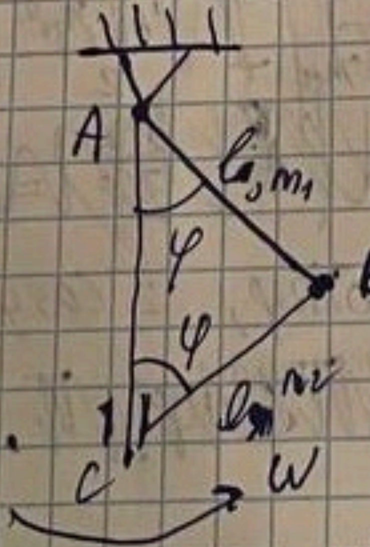
$AB = l$ m_1

$BC = l$ m_2

ω, $\varphi = ?$

ПР

Решение:



$$T_{отн} = \frac{1}{2}\frac{m_1 l^2}{3}\dot\varphi^2 + \frac{1}{2}\frac{m_2 l^2}{3}\dot\varphi^2 = \frac{1}{6}l^2\dot\varphi^2(m_1+m_2)$$

$$T_{абс} = \frac{m_1 v_{абс1}^2}{2} + \frac{m_2 v_{абс2}^2}{2} = \frac{m_1(v_{отн1}+v_{пер1})^2}{2} + \frac{m_2(v_{отн2}+v_{пер2})^2}{2} =$$

$$= \frac{m_1 v_{отн1}^2}{2} + \frac{m_2 v_{отн2}^2}{2} + \frac{m_1 v_{пер1}^2}{2} + \frac{m_2 v_{пер2}^2}{2} + \overset{0}{\cancel{(v_{отн}, v_{пер})}}(m_1+m_2) =$$

$$= \frac{1}{6}l^2\dot\varphi^2(m_1+m_2) + \frac{1}{2}\cdot\frac{1}{3}m_1 l^2\sin^2\varphi\,\omega^2 + \frac{1}{6}m_2 l^2\sin^2\varphi\,\omega^2 = \frac{1}{6}l^2\dot\varphi^2(m_1+m_2) + \frac{1}{6}l^2\sin^2\varphi\,\omega^2(m_1+m_2)$$

$$V = T_{отн} - T_{абс} = -\frac{1}{6}l^2\sin^2\varphi\,\omega^2(m_1+m_2)$$

$$Q_{ин} = \frac{d}{dt}\frac{\partial V}{\partial\dot\varphi} - \frac{\partial V}{\partial\varphi} = \cancel{\frac{1}{6}}\frac{1}{3}l^2\sin\varphi\cos\varphi\,\omega^2(m_1+m_2)$$

$Q_{пот} = -\frac{\partial П}{\partial \varphi}$, $П = -\frac{1}{2} m_1 l \cos\varphi\, g - m_2 g \left(l\cos\varphi + \frac{1}{2} l \cos\varphi\right)$

$-\frac{\partial П}{\partial \varphi} = -\frac{1}{2} m_1 l \sin\varphi\, g - m_2 g l \cdot \frac{3}{2}\sin\varphi = -gl\sin\varphi\left(\frac{1}{2}m_1 + \frac{3}{2}m_2\right)$

$Q_{ин} + Q_{пот} = 0$

$-gl\sin\varphi\left(\frac{1}{2}m_1 + \frac{3}{2}m_2\right) + \frac{1}{3} l^2 \sin\varphi\cos\varphi\,\omega^2(m_1+m_2) = 0$

$-g\left(\frac{1}{2}m_1 + \frac{3}{2}m_2\right) + \frac{1}{3} l\cos\varphi\,\omega^2(m_1+m_2) = 0$ или $\sin\varphi = 0$

$\cos\varphi = +\frac{3g(m_1+3m_2)}{2l\omega^2(m_1+m_2)} \leq 1 \geq -1$

$$\left[\begin{array}{l} \varphi_1 = 0 \\ \varphi_2 = \arccos\left(\frac{+3g(m_1+3m_2)}{2l\omega^2(m_1+m_2)}\right), \quad \omega^2 \geq \frac{3g(m_1+3m_2)}{2l(m_1+m_2)} \end{array}\right.$$

№17.11

д) $\ddot{x} + \dot{x} + x - \alpha y = 0$, $\bar{q} = (x, y)^T$

$\beta\dot{y} - x + y = 0$

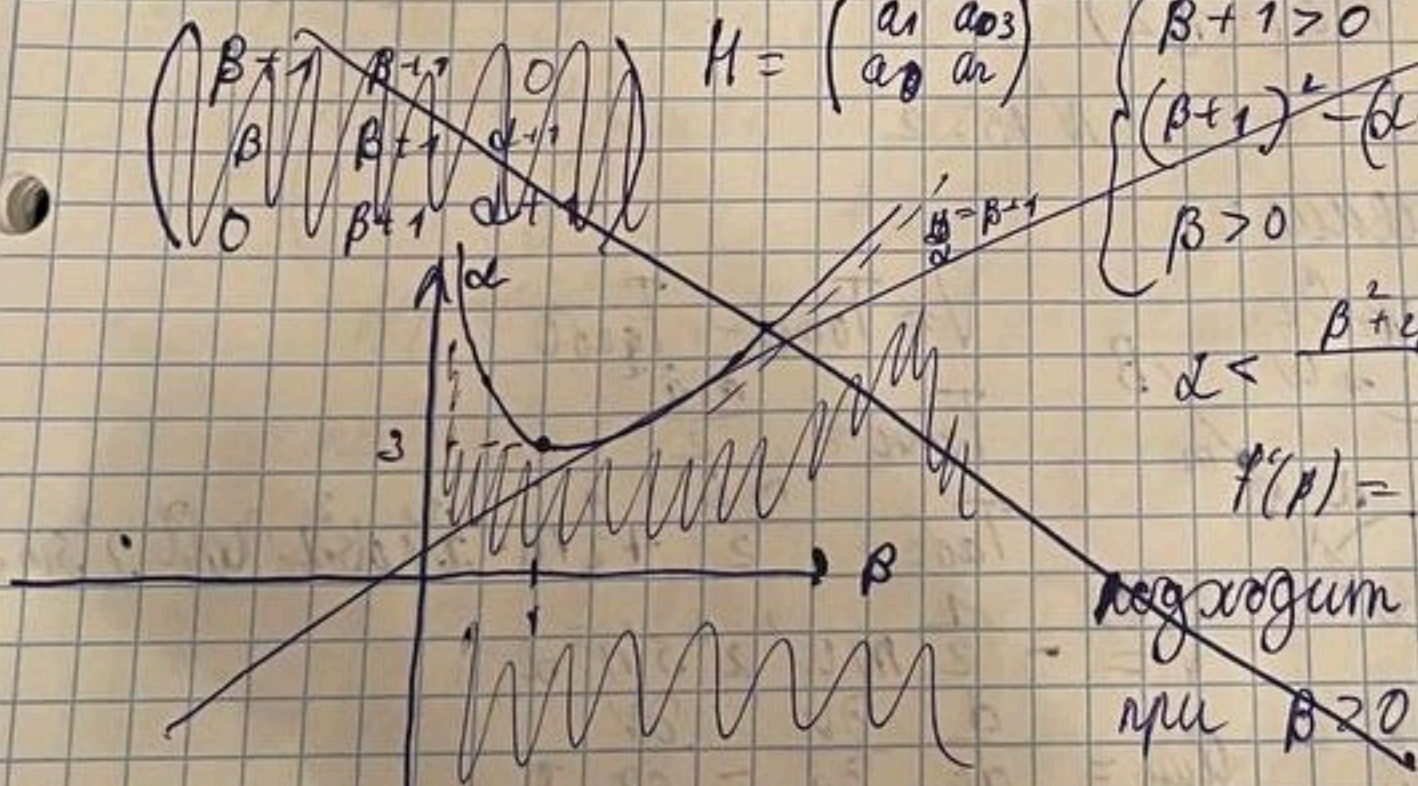
$A\ddot{q} + B\dot{q} + Cq = 0$, $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ 0 & \beta \end{pmatrix}$, $C = \begin{pmatrix} 1 & -\alpha \\ -1 & 1 \end{pmatrix}$

$A\lambda^2 + B\lambda + C = 0$

$\det(A\lambda^2 + B\lambda + C) = 0$ $\quad \begin{vmatrix} \lambda^2+\lambda+1 & -\alpha \\ -1 & \beta\lambda+1 \end{vmatrix} = 0$

$(\lambda^2+\lambda+1)(\beta\lambda+1) + \alpha = 0$ $\quad \underset{a_0}{\beta}\lambda^3 + \underset{a_1}{(\beta+1)}\lambda^2 + \underset{a_2}{(\beta+1)}\lambda + \underset{a_3}{\alpha+1} = 0$

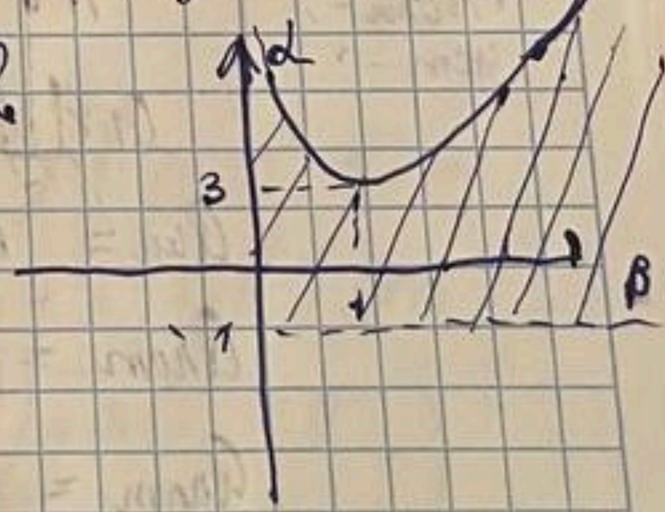
$H = \begin{pmatrix} a_1 & a_3 \\ a_0 & a_2 \end{pmatrix}$ $\quad \begin{cases} \beta+1>0 \\ (\beta+1)^2 - (\alpha+1)\beta > 0 \\ \beta > 0 \end{cases}$ $\quad \begin{array}{l} \beta > -1 \\ \frac{(\beta+1)^2}{\beta} - 1 > \alpha \\ \beta > 0 \end{array}$



$\alpha < \frac{\beta^2+2\beta+1-\beta}{\beta} = \beta + 1 + \frac{1}{\beta}$

$f'(\beta) = 1 - \frac{1}{\beta^2} = 0$, $\beta = 1$

подходит все, что под графиком при $\beta > 0$



$H = \begin{pmatrix} a_1 & a_0 & 0 \\ a_3 & a_2 & a_1 \\ 0 & 0 & a_3 \end{pmatrix} = \begin{pmatrix} \beta+1 & \beta & 0 \\ \alpha+1 & \beta+1 & \beta+1 \\ 0 & 0 & \alpha+1 \end{pmatrix}$ ~~det~~

$\begin{cases} \beta > 0 \\ \beta+1 > 0 \\ (\beta+1)^2 - \beta(\alpha+1) > 0 \\ (\alpha+1)\left((\beta+1)^2 - \beta(\alpha+1)\right) > 0 \end{cases}$ $\quad \begin{cases} \alpha > -1 \\ \beta > 0 \\ (\beta+1)^2 - \beta(\alpha+1) > 0 \end{cases}$ $\Rightarrow$ $\beta^2 + \beta + 1 > \alpha\beta$

2-б) $\ddot{x}+\dot{x}+x-\alpha y=0$ $A=\begin{pmatrix}1&0\\0&0\end{pmatrix},\ B=\begin{pmatrix}1&0\\0&1\end{pmatrix},\ C=\begin{pmatrix}1&-\alpha\\-\beta&1\end{pmatrix}$

$\dot{y}-\beta x+y=0$ $q=(x,y)^T$

$A\ddot{q}+B\dot{q}+C=0$

$A\lambda^2+B\lambda+C=0$ $\det(A\lambda^2+B\lambda+C)=0$

$$\begin{vmatrix}\lambda^2+\lambda+1 & -\alpha\\ -\beta & \lambda+1\end{vmatrix}=\lambda^3+1+2\lambda^2+2\lambda+\alpha\beta=$$

$$=\overset{a_0}{\lambda^3}+\overset{a_1}{2\lambda^2}+\overset{a_2}{2\lambda}+\overset{a_3}{\alpha\beta+1}=0$$

$$\begin{pmatrix}2&1&0\\ \alpha\beta+1&2&2\\ 0&0&\alpha\beta+1\end{pmatrix}$$

$\Delta_1=2>0$

$\Delta_2=4-(\alpha\beta+1)>0$ $\alpha\beta<3$

$\Delta_3=(\alpha\beta+1)(3-\alpha\beta)>0$ $\alpha\beta>-1$

Ответ: $-1<\alpha\beta<3$

№17.31

$f(\lambda)=\lambda^4+\lambda^3+5\lambda^2+(2+k)\lambda+4=0$

$$\begin{pmatrix}1&2+k&0&0\\1&5&4&0\\0&1&2+k&0\\0&1&5&4\end{pmatrix}$$

$\Delta_1>0$

$\Delta_2=5-2-k>0$ $k<3$

$\Delta_3=(3-k)(2+k)-4>0$ $-k^2+k+2>0$

$\Delta_4=4((3-k)(2+k)-4)>0$ $k^2-k-2<0$

$(k-2)(k+1)<0$

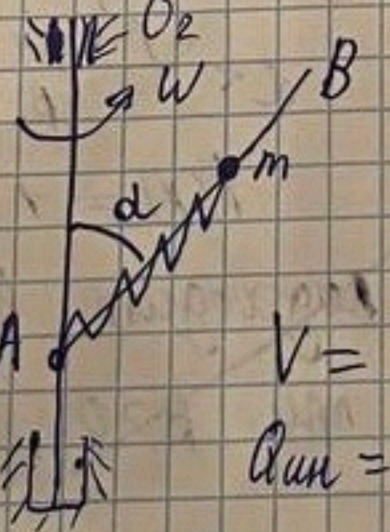
$k\in(-1,2)$

Ответ: $k\in(-1,2)$

№15.2

Дано:
$AB,\ \alpha$
$\omega,\ m$
$c,\ l_0$
$ПР_{отн}$ - ?
уст - ?

Решение:



$V=T_{отн}-T_{абс}$

$T_{отн}=\dfrac{m\dot{x}^2}{2}$

$T_{абс}=\dfrac{m\dot{x}^2}{2}+\dfrac{1}{2}m\cdot x^2\sin^2\alpha\,\omega^2$

$V=-\dfrac{1}{2}m\omega^2x^2\sin^2\alpha$

$Q_{ин}=\dfrac{d}{dt}\dfrac{\partial V}{\partial\dot{x}}-\dfrac{\partial V}{\partial x}$

$Q_{ин}=m\omega^2x\sin^2\alpha$

$Q_{пот}=-\dfrac{\partial П}{\partial x}$; $П=\dfrac{c(x-l_0)^2}{2}+mgx\cos\alpha$

$Q_{пот}=-c(x-l_0)-mg\cos\alpha$

$Q_{ин}+Q_{пот}=0$ $m\omega^2x\sin^2\alpha-c(x-l_0)-mg\cos\alpha=0$

$x=\dfrac{mg\cos\alpha-cl_0}{m\omega^2\sin^2\alpha-c}$ — ПР

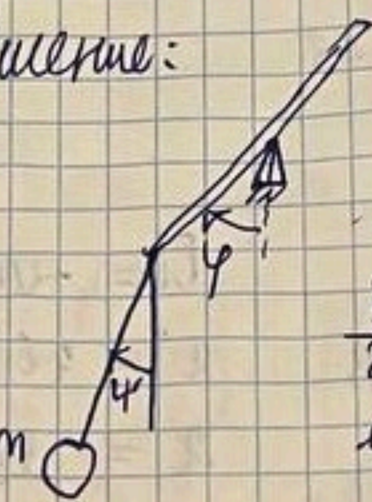
$-\frac{\partial^2(V+\Pi)}{\partial x^2} = +\frac{\partial(Q_{ин}+Q_{пот})}{\partial x} = m\omega^2\sin^2\alpha - c > 0$ неустойчиво

$\Rightarrow c > m\omega^2\sin^2\alpha$ – устойчиво.

N 15.13

Дано:
m, l, M
$AO = l_1,\ OB = l_2$
ПР – ?
устойч.

Решение:



$\Pi = M\cdot\left(\frac{l_2+l_1}{2} - l_1\right)\cos\varphi -$
$-(l_1\cos\varphi + l\cos\psi)m$

$\frac{\partial\Pi}{\partial\varphi} = M\sin\varphi\left(l_1 - \frac{l_2+l_1}{2}\right) +$
$l_1\sin\varphi\, m = 0$

$\frac{\partial\Pi}{\partial\psi} = +ml\sin\psi = 0,\ \left[\psi = 0\right.$ $\quad \left[\begin{array}{l}\varphi = 0,\ \varphi = \pi \\ M\frac{l_1-l_2}{2} + ml_1 = 0\end{array}\right.$

Если $2ml_1 = M(l_2-l_1)$, то ПР – бесс. мн-во, $\psi = 0$, φ – любое. Иначе: $\psi = 0, \varphi = 0$ или $\varphi = \pi$

$\cancel{\frac{\partial^2\Pi}{\partial^2\varphi} = M\cos\varphi\left(l_1 - \frac{l_2+l_1}{2}\right) + l_1\cos\varphi\, m}$

$\cos\varphi \approx 1 - \frac{\varphi^2}{2}$ $\quad$ | $\Pi_2 = M\left(\frac{l_2-l_1}{2}\right)\left(1-\frac{\varphi^2}{2}\right) - \left(l_1\left(1-\frac{\varphi^2}{2}\right) +\right.$

$\sin\varphi \approx \varphi,\ \sin\psi \approx \psi$ $\quad$ | $\left. + l\left(1-\frac{\psi^2}{2}\right)\right)m$

$C = \begin{pmatrix} -M\left(\frac{l_2-l_1}{2}\right) + ml_1 & 0 \\ 0 & +ml \end{pmatrix}$ $\quad$ $\Delta 1 = M\left(\frac{l_1-l_2}{2}\right) + ml_1$
$\Delta 2 = ml\,\Delta 1$

Если $M\left(\frac{l_1-l_2}{2}\right) + ml_1 > 0$ – $2ml_1 > M(l_2-l_1)$, то ПР – устойчиво ($\varphi = 0, \psi = 0$). Если меньше – неустойчиво.

~~Случаи $\varphi = 0, \psi = \pi$, $\varphi = \pi, \psi = 0$, $\varphi = \pi, \psi = \pi$ тоже НЛ, но не~~

$\varphi = \pi,\ \psi = 0$ $\quad$ $\Pi_2 = M\left(\frac{l_2-l_1}{2}\right)\left(\frac{1}{2}(\varphi-\pi)^2 - 1\right) -$
$-\left(l_1\frac{1}{2}\left((\varphi-\pi)^2 - 1\right) + l\left(1-\frac{\psi^2}{2}\right)\right)m$

$\cos\varphi \approx -1 + (\varphi-\pi)^2\cdot\frac{1}{2}$

$\sin\varphi \approx \varphi - \pi$

$\cos\psi \approx 1 - \frac{\psi^2}{2}$

$C = \begin{pmatrix} M\left(\frac{l_2-l_1}{2}\right) - ml_1 & 0 \\ 0 & ml \end{pmatrix}$

$\Delta 1 = M\left(\frac{l_2-l_1}{2}\right) - ml_1 > 0$ $\quad 2ml_1 < M(l_2-l_1)$ – $\varphi = \pi, \psi = 0$ – уст.,

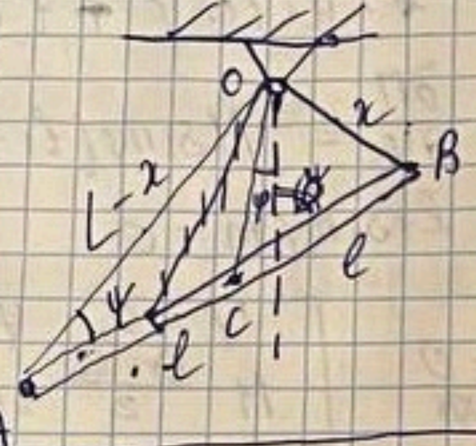
$2ml_1 > M(l_2-l_1)$ – неустойчиво

Ответ: $\varphi=0, \psi=0$ – ПР, $2ml_1 > M(l_2-l_1)$ – устойч., $2ml_1 < M(l_2-l_1)$ неуст. $\varphi=\pi, \psi=0$ – ПР, $2ml_1 < M(l_2-l_1)$ уст., $2ml_1 > M(l_2-l_1)$ неуст. Если $M(l_2-l_1)=2ml_1$, то ПР беск. много φ – любое, $\psi=0$

№15.28

Дано:
$2l, m$
$L>2l$
ПР – ?
уст – ?

Решение:



$\cancel{\Pi = -Am}$

$$x^2 = 4l^2 + (L-x)^2 - 4l(L-x)\cos\psi$$

$$x = \frac{4l^2+L^2-4lL\cos\psi}{2L-4l\cos\psi}$$

$$AO = \frac{L^2-4l^2}{2L-4l\cos\psi}$$

$$OC = \frac{\sqrt{L^4+16l^4-4L^2l^2-4L^3l\cos\psi+8L^2l\cos\psi-16l^4\cos^2\psi}}{2(L-2l\cos\psi)}$$

$$\Pi = -mg\,OC(\psi)\cdot\cos\varphi, \quad \frac{\partial\Pi}{\partial\varphi} = -mg\,OC'\cos\varphi + mg\,OC\sin\varphi = 0$$

$\psi = 0, \pi$ – ПР

$\cancel{\frac{\partial\Pi}{\partial\varphi} = mg\,OC\sin\varphi = 0 \quad \varphi = 0, \pi}$

$\cancel{\frac{\partial^2\Pi}{\partial\varphi^2} = mg\,OC\cos\varphi}$

№16.17

Дано:
m, $2ly = x^2$
$T = T_{мат}$ – пок-ть

Решение:

$$\Pi = \cancel{mg\sqrt{\ldots}}\; mgy = mg\frac{x^2}{2l}$$

$$\frac{\partial\Pi}{\partial x} = \frac{1}{l}mgx, \quad x=0, y=0 \text{ – ПР}$$

$$\cancel{\frac{\partial^2\Pi}{\partial x^2} = mg} \quad F_x = -\frac{\partial\Pi}{\partial x} = -mgx\,l$$

$$m\ddot{x} = -mgx\cdot\frac{1}{l}$$

$$\ddot{x} + \frac{g}{l}x = 0 \text{ – уравнение колебаний}$$

$$\omega^2 = \frac{g}{l}, \quad T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{l}{g}}$$

$$T_{мат} = 2\pi\sqrt{\frac{l}{g}} = T \text{ – показали}$$

№16.40

Дано:

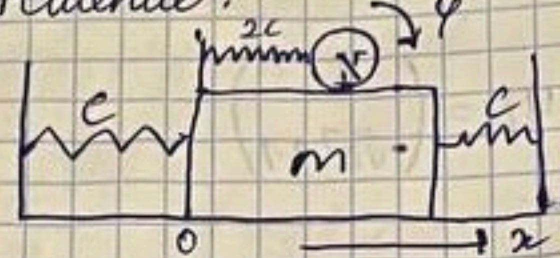
m, c

$\frac{m}{2}$, r

$2c$

колеб - ?

Решение:



$\Pi = c x^2 + 2c \cdot r^2 \varphi^2$

$\frac{\partial \Pi}{\partial x} = 2cx$, $x = 0$, $\frac{\partial \Pi}{\partial (r\varphi)} = 2cr\varphi$, $r\varphi = 0$

$x = 0$, $r\varphi = 0$ – ПР

$C = \begin{pmatrix} 2c & 0 \\ 0 & 2c \end{pmatrix}$

$T = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} \cdot \frac{m}{2} (\dot{x} + r\dot{\varphi})^2 = \frac{3}{4} m \dot{x}^2 + \frac{1}{4} m r^2 \dot{\varphi}^2 + \frac{1}{2} m \dot{x} r \dot{\varphi}$

$A = \begin{pmatrix} \frac{3}{2}m & \frac{1}{2}m \\ \frac{1}{2}m & \frac{1}{2}m \end{pmatrix}$

$A\ddot{q} + Cq = 0$, $q = (x, r\varphi)^T$

$\det(C - \lambda A) = 0$

$\begin{vmatrix} 2c - \lambda \frac{3}{2} m & \frac{1}{2}\lambda m \\ \frac{1}{2}\lambda m & 2c - \frac{1}{2}\lambda m \end{vmatrix} = 4c^2 - 4c\lambda m + \frac{3}{4}\lambda^2 m^2 - \frac{1}{4}\lambda^2 m^2 = 0$

$\cancel{4c^2 - \frac{7}{2} c\lambda m + \frac{1}{8}\lambda^2 m^2 = 0}$

$\cancel{32c^2 - 28c\lambda m + \lambda^2 m^2 = 0}$

$\cancel{D = 784c^2m^2 - 128c^2m^2 = 656c^2m^2}$

$\cancel{= \frac{14c \pm 2\sqrt{41}c}{m}}$

$\cancel{(C - \lambda_1 A)u_1 = 0}$

$4c^2 - 4c\lambda m + \frac{1}{2}\lambda^2 m^2 = 0$

$8c^2 - 8c\lambda m + \lambda^2 m^2 = 0$

$m^2\lambda^2 - 8cm \cdot \lambda + 8c^2 = 0$ $D = 64c^2m^2 - 32c^2m^2 = 32c^2m^2$

$\lambda = \frac{8cm \pm 4\sqrt{2}cm}{2m^2} = \frac{4c \pm 2\sqrt{2}c}{m} = \left[\begin{array}{l} \frac{c}{m}(4 + 2\sqrt{2}) \\ \frac{c}{m}(4 - 2\sqrt{2}) \end{array} \right.$

$(C - \lambda_1 A)u_1 = 0$

$\begin{pmatrix} 2c - \frac{3}{2}c(4 + 2\sqrt{2}) & \frac{1}{2}c(4 + 2\sqrt{2}) \\ \frac{1}{2}c(4 + 2\sqrt{2}) & 2c - \frac{c}{2}(4 + 2\sqrt{2}) \end{pmatrix} \to \begin{pmatrix} -3\sqrt{2} - 4 & 2 + \sqrt{2} \\ 2 + \sqrt{2} & -\sqrt{2} \end{pmatrix}$

$(2 + \sqrt{2}) \cdot \left(\frac{2 + \sqrt{2}}{\sqrt{2}}\right) = \frac{4 + 2 + 4\sqrt{2}}{\sqrt{2}} = 4 + 3\sqrt{2}$ | $(2 + \sqrt{2} \quad -\sqrt{2})$ $u_1 = \begin{pmatrix} \sqrt{2} & 1 \\ & 1 + \sqrt{2} \end{pmatrix}$

$(C - \lambda_2 A) u_2 = 0 \quad \begin{pmatrix} 2c - \frac{3}{2}c(4-2\sqrt{2}) & \frac{1}{2}c(4-2\sqrt{2}) \\ \frac{1}{2}c(4-2\sqrt{2}) & 2c - \frac{1}{2}c(4-2\sqrt{2}) \end{pmatrix}$

$\begin{pmatrix} -4+3\sqrt{2} & 2-\sqrt{2} \\ 2-\sqrt{2} & \sqrt{2} \end{pmatrix} \quad u_2 = \begin{pmatrix} 1 \\ -\sqrt{2}+1 \end{pmatrix}$

$\begin{pmatrix} x \\ r\varphi \end{pmatrix} = b_1 \begin{pmatrix} 1 \\ 1+\sqrt{2} \end{pmatrix} \sin\left(\sqrt{\frac{c}{m}(4+2\sqrt{2})}\, t + \alpha_1\right) + b_2 \begin{pmatrix} 1 \\ 1-\sqrt{2} \end{pmatrix} \sin$

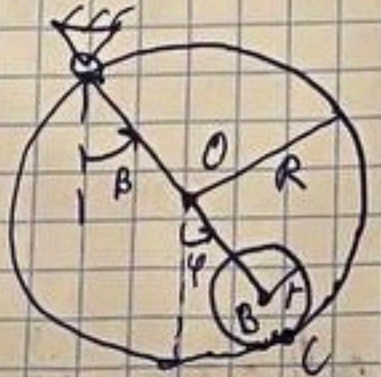
$\cdot \sin\left(\sqrt{\frac{c}{m}(4-2\sqrt{2})} + \alpha_2\right),$

Ответ: $\begin{pmatrix} x \\ r\varphi \end{pmatrix} = b_1 \begin{pmatrix} 1 \\ 1+\sqrt{2} \end{pmatrix} \sin\left(\sqrt{\frac{c}{m}(4+2\sqrt{2})}\, t + \alpha_1\right) + b_2 \begin{pmatrix} 1 \\ 1-\sqrt{2} \end{pmatrix} \sin\left(\sqrt{\frac{c}{m}(4-2\sqrt{2})}\, t + \alpha_2\right)$

N 16.51

Дано:
$R,\ m$
$r = \frac{R}{4},\ m$
колеб.-?

Решение:



$\Pi = -mgR\cos\beta - mg\left(R\cos\beta + (R-r)\cos\varphi\right) = - 8mgr\cos\beta - 3rmg\cos\varphi$

$\frac{\partial \Pi}{\partial \varphi} = 3rmg\sin\varphi,\quad \varphi = 0 - \Pi P,\quad \frac{\partial \Pi}{\partial \beta} = 8mgr\sin\beta,\quad \beta = 0 - \Pi P$

$\Pi_2 = -8mgr\left(1 - \frac{\beta^2}{2}\right) - 3rmg\left(1 - \frac{\varphi^2}{2}\right)$

$\frac{\partial^2 \Pi}{\partial^2 \varphi} = 3mgr,\quad \frac{\partial^2 \Pi}{\partial^2 \beta} = 8mgr,\quad \frac{\partial^2 \Pi}{\partial \varphi \partial \beta} = 0$

$C = \begin{pmatrix} 8mgr & 0 \\ 0 & 3mgr \end{pmatrix}$

$T = \frac{1}{2} m \cdot \dot\beta^2 R^2 \cdot 2 + \frac{1}{2} m v_{цм}^2 + \frac{1}{2} J\omega^2$

$\bar v_0 = \bar\omega \times \bar R = \begin{pmatrix} 0 \\ 0 \\ \dot\beta \end{pmatrix} \times \begin{pmatrix} R\sin\beta \\ -R\cos\beta \\ 0 \end{pmatrix} = \begin{pmatrix} R\cos\beta\,\dot\beta \\ -R\sin\beta\,\dot\beta \\ 0 \end{pmatrix}$

$\bar v_{цм} = \bar v_0 + \omega \times \bar{O_{цм}} = \begin{pmatrix} R\cos\beta\,\dot\beta \\ -R\sin\beta\,\dot\beta \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot\varphi \end{pmatrix} \times \begin{pmatrix} (R-r)\sin\varphi \\ -(R-r)\cos\varphi \\ 0 \end{pmatrix}$

$= \begin{pmatrix} R\cos\beta\,\dot\beta + (R-r)\cos\varphi\,\dot\varphi \\ -R\sin\beta\,\dot\beta - (R-r)\sin\varphi\,\dot\varphi \\ 0 \end{pmatrix} = \begin{pmatrix} 4r\cos\beta\,\dot\beta + 3r\cos\varphi\,\dot\varphi \\ -4r\sin\beta\,\dot\beta - 3r\sin\varphi\,\dot\varphi \\ 0 \end{pmatrix}$

$\cancel{v_{ум}^2 = 16r^2\cos^2\beta\dot\beta^2 + 9r^2\cos^2\varphi\dot\varphi^2 + \ldots}$

$\cancel{+ 16r^2\sin^2\beta\dot\beta^2 + 9r^2\sin^2\beta\dot\beta^2 + 24r^2\sin\beta\sin\varphi\dot\beta\dot\varphi = }$

$\cancel{= 16r^2\dot\beta^2 + 9r^2\dot\varphi^2 + 24r^2\dot\beta\dot\varphi}$

$\cancel{v_{ум} = (4r\dot\beta^2}$

В ПР $\bar v_{ум} = \begin{pmatrix} 4r\dot\beta + 3r\dot\varphi \\ 0 \\ 0 \end{pmatrix}$, $v_{ум} = 4r\dot\beta + 3r\dot\varphi$

$\bar v_{B\,отн} = \bar v_C + \bar\omega_{CB} \times \bar{CB}$ $\quad \begin{pmatrix} 4r\dot\beta + 3r\dot\varphi \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \omega_{CB} \end{pmatrix} \times \begin{pmatrix} 0 \\ -r \\ 0 \end{pmatrix}$

(B - центр масс)

$4r\dot\beta + 3r\dot\varphi = \omega_{CB}\cdot r$, $\omega_{CB} = 4\dot\beta + 3\dot\varphi$

$T = 2m\dot\beta^2 R^2 + \frac{1}{2}m(4r\dot\beta + 3r\dot\varphi)^2 + \frac{1}{4}mr^2 \cdot 9\dot\varphi^2 =$

$= 16m\dot\beta^2 r^2 + \frac{1}{2}m(4r\dot\beta + 3r\dot\varphi)^2 + \frac{9}{4}mr^2\dot\varphi^2$

$A = \begin{pmatrix} 48mr^2 & 12mr^2 \\ 12mr^2 & \frac{27}{2}mr^2 \end{pmatrix} = mr^2\begin{pmatrix} 32 & 12 \\ 12 & \frac{27}{2} \end{pmatrix}$

$\det(C - \varkappa A) = 0$

$\begin{vmatrix} 8mgr - 48\varkappa mr^2 & -12mr^2\varkappa \\ -12mr^2\varkappa & 3mgr - \frac{27}{2}mr^2\varkappa \end{vmatrix} = 24m^2g^2r^2 + 648m^2r^4\varkappa^2 - 252m^2r^3g\varkappa - 144m^2r^4\varkappa^2 = 0$

$\cancel{12m^2r^4\varkappa^2\ 24m^2r^4\varkappa^2 - 17m^2r^3g\varkappa + 2m^2g^2r^2 = 0}$

$\cancel{24r^2\varkappa^2 - 17rg\varkappa + 2g^2 = 0;\ D = 289r^2g^2 - 192r^2g^2 =}$

$504m^2r^4\varkappa^2 - 252m^2r^3g\varkappa + 24m^2g^2r^2 = 0$

$42r^2\varkappa^2 - 21rg\varkappa + 2g^2 = 0 \quad D = 441r^2g^2 - 336r^2g^2 = 105r^2g^2$

$\varkappa = \frac{21rg \pm \sqrt{105}\,rg}{84r^2} = \frac{g}{r}\left(\frac{1}{4} \pm \frac{\sqrt{105}}{84}\right) \approx$

$\approx \frac{g}{r}\left(\frac{1}{4} \pm \frac{1}{8}\right) = \left[\begin{array}{l} \frac{3}{8}\cdot\frac{g}{r} \\ \frac{1}{8}\frac{g}{r} \end{array}\right.$

$\left(-12mr^2\cdot\frac{g}{r}\frac{9}{64} \quad 3mgr - \frac{27}{2}mr^2\frac{g}{r}\frac{9}{64}\right)u_1 = 0$

$\left(-\frac{27}{64} \quad \frac{141}{128}\right)$ $u_1 = \begin{pmatrix} 47/2 \\ 9 \end{pmatrix}$, $u_1 = \begin{pmatrix} 47 \\ 18 \end{pmatrix}$

$\left(-12mr^2\cdot\frac{g}{r}\frac{1}{64} \quad 3mgr - \frac{27}{2}mr^2\frac{g}{r}\frac{1}{64}\right)$, $\left(-\frac{3}{16} \quad \frac{357}{128}\right)$ $u_2 = \begin{pmatrix} 119 \\ 8 \end{pmatrix}$

$$\begin{pmatrix}\beta\\ \varphi\end{pmatrix} = b_1\begin{pmatrix}47\\ 18\end{pmatrix}\sin\left(\sqrt{\frac{3}{8}\frac{g}{r}}\,t+\alpha_1\right)+b_2\begin{pmatrix}119\\ 8\end{pmatrix}\sin\left(\sqrt{\frac{2}{8}\frac{g}{r}}\,t+\alpha_2\right)$$

N16.70

$q = (x_1, x_2, \varphi_1, \varphi_2)$

Дано:

m, c

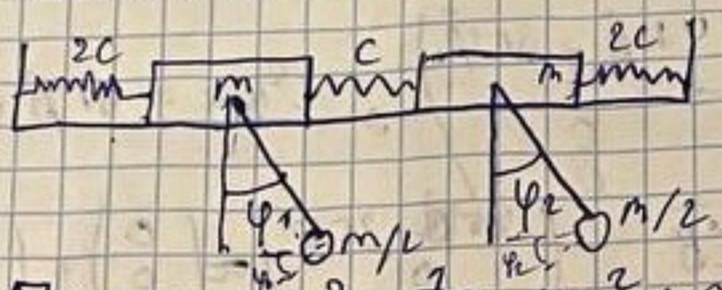
$2c$

$\frac{m}{2}, l$

$2cl = mg$

колеб-?

Решение:



$$\Pi = \frac12 2c\cdot x_1^2 + \frac12 2c x_2^2 + \frac12 c(x_1+x_2)^2 - \frac12 mgl\cos\varphi_1 - \frac12 mgl\cos\varphi_2 = cx_1^2 + cx_2^2 +$$

$$+\frac12 cx_1^2 + \frac12 cx_2^2 + cx_1x_2 - \frac12 mgl(\cos\varphi_1+\cos\varphi_2)$$

$$= \frac32 cx_1^2 + \frac32 cx_2^2 + cx_1x_2 - \frac12 mgl(\cos\varphi_1+\cos\varphi_2)$$

$$\frac{\partial\Pi}{\partial x_1} = 3cx_1 + cx_2 = 0 \quad x_1 = x_2 = 0 - \Pi P$$

$$\frac{\partial\Pi}{\partial x_2} = 3cx_2 + cx_1 = 0$$

$$\frac{\partial\Pi}{\partial\varphi_1} = +\frac12 mgl\sin\varphi_1, \ \varphi_1 = 0 \quad - \Pi P$$

$$\frac{\partial\Pi}{\partial\varphi_2} = +\frac12 mgl\sin\varphi_2, \ \varphi_2 = 0$$

$$\Pi_2 = \frac32 cx_1^2 + \frac32 cx_2^2 + cx_1x_2 - \frac12 mgl\left(1-\frac{\varphi_1^2}{2}+1-\frac{\varphi_2^2}{2}\right)$$

$$\frac{\partial^2\Pi_2}{\partial x_1^2} = 3c, \quad \frac{\partial^2\Pi_2}{\partial x_2^2} = 3c, \quad \frac{\partial^2\Pi_2}{\partial\varphi_1^2} = +\frac12 mgl, \quad \frac{\partial^2\Pi_2}{\partial\varphi_2^2} = \frac12 mgl$$

$$\frac{\partial^2\Pi_2}{\partial x_1\partial x_2} = c, \quad \frac{\partial^2\Pi_2}{\partial\varphi_1\partial\varphi_2} = 0$$

$$C = \begin{pmatrix} 3c & c & 0 & 0\\ c & 3c & 0 & 0\\ 0 & 0 & 1/2mgl & 0\\ 0 & 0 & 0 & 1/2mgl \end{pmatrix}$$

$$\cancel{T = \frac12 m\dot x_1^2 + \frac12 m\dot x_2^2 + \frac14 m\dot\varphi_1^2 l^2 + \frac14 m\dot\varphi_2^2 l^2}$$

$$\cancel{A = \begin{pmatrix} m & 0 & 0 & 0\\ 0 & m & 0 & 0\\ 0 & 0 & \frac12 ml^2 & 0\\ 0 & 0 & 0 & \frac12 ml^2 \end{pmatrix}} \qquad \cancel{\det(C-\lambda A) = 0}$$

$$\begin{vmatrix} 3c-\lambda m & c & 0 & 0 \\ c & 3c-\lambda m & 0 & 0 \\ 0 & 0 & 1/2\,mgl-\frac{1}{2}\lambda ml^2 & 0 \\ 0 & 0 & 0 & 1/2\,mgl-\frac{1}{2}\lambda ml^2 \end{vmatrix} = \left(\frac{1}{2}mgl-\frac{1}{2}\lambda ml^2\right)^2\cdot\left((3c-\lambda m)^2-c^2\right)=0$$

$$\left(\frac{1}{2}g-\lambda l\right)^2\left((3c-\lambda m)^2-c^2\right)=0 \qquad \lambda_{1,2}=\frac{g}{2l}$$

$$3c-\lambda m=\pm c \qquad \lambda_3=\frac{2c}{m},\ \lambda_4=\frac{4c}{m}$$

$$2cl=mg,\ \Rightarrow\ \frac{2c}{m}=\frac{g}{l}$$

$$\lambda_3=\frac{g}{l},\ \lambda_4=\frac{2g}{l}$$

$$\lambda_1=\frac{g}{l}=\frac{2c}{m} \quad \begin{pmatrix} c & c & 0 & 0 \\ c & c & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad u_1=\begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix},\ \lambda_2=\frac{g}{l} \quad u_2=\begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}$$

$$\lambda_3=\frac{g}{l} \quad u_3\begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}$$

$$\lambda_4=\frac{2g}{l}=\frac{4c}{m} \quad \begin{pmatrix} -c & c & 0 & 0 \\ c & -c & 0 & 0 \\ 0 & 0 & -\frac{1}{2}mgl & 0 \\ 0 & 0 & 0 & -\frac{1}{2}mgl \end{pmatrix} \quad u_4=\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$T=\frac{1}{2}m\dot{x}_1^2+\frac{1}{2}m\dot{x}_2^2+T_{шат}$$

$$T_{шат}=\frac{1}{4}m(\dot{x}_1+\dot{\varphi}_1 l)^2+\frac{1}{4}m(\dot{x}_2+\dot{\varphi}_2 l)^2=\frac{1}{4}m\dot{x}_1^2+\frac{1}{4}m\dot{\varphi}_1^2l^2+$$
$$+\frac{1}{2}m\dot{x}_1\dot{\varphi}_1 l\cdot\cos\varphi_1+\frac{1}{4}m\dot{x}_2^2+\frac{1}{4}m\dot{\varphi}_2^2l^2+\frac{1}{2}m\dot{x}_2\dot{\varphi}_2 l\cos\varphi_2=$$
$$=\frac{1}{4}m\dot{x}_1^2+\frac{1}{4}m\dot{\varphi}_1^2l^2+\frac{1}{2}m\dot{x}_1\dot{\varphi}_1 l+\frac{1}{4}m\dot{x}_2^2+\frac{1}{4}m\dot{\varphi}_2^2l^2+\frac{1}{2}m\dot{x}_2\dot{\varphi}_2 l$$

$$A=\begin{pmatrix} \frac{3}{2}m & 0 & \frac{1}{2}ml & 0 \\ 0 & \frac{3}{2}m & 0 & \frac{1}{2}ml \\ \frac{1}{2}ml & 0 & \frac{1}{2}ml^2 & 0 \\ 0 & \frac{1}{2}ml & 0 & \frac{1}{2}ml^2 \end{pmatrix} \qquad \det(C-\lambda A)=0$$

$$\begin{vmatrix} 3c-\frac{3}{2}m\lambda & c & -\frac{1}{2}ml\lambda & 0 \\ c & 3c-\frac{3}{2}\lambda m & 0 & -\frac{1}{2}ml\lambda \\ -\frac{1}{2}ml\lambda & 0 & \frac{1}{2}mgl-\frac{1}{2}ml^2\lambda & 0 \\ 0 & -\frac{1}{2}ml\lambda & 0 & \frac{1}{2}mgl-\frac{1}{2}ml^2\lambda \end{vmatrix} = \left(\frac{1}{2}mgl-\frac{1}{2}ml\lambda\right)^2\cdot$$
$$\cdot\left(\left(3c-\frac{3}{2}\lambda m\right)^2-c^2\right)+$$
$$+\left(\frac{1}{2}mgl-\frac{1}{2}ml\lambda\right)\cdot$$
$$\cdot\left(-\frac{1}{2}ml\lambda\right)\cdot\left(3c-\frac{3}{2}\lambda m\right)+$$
$$+\ -\frac{1}{2}ml\lambda\left(-\frac{1}{2}ml\lambda\left(\left(3c-\frac{3}{2}m\lambda\right)\cdot\left(\frac{1}{2}mgl-\frac{1}{2}ml\lambda\right)-\right.\right.$$
$$\left.\left.-\left(\frac{1}{2}ml\lambda\right)^2\right)\right)=\frac{1}{4}\left(m^2g^2l^2+m^2l^2\lambda^2-2m^2l^2g\lambda\right)\left(8c^2+\frac{9}{4}\lambda^2m^2-9\lambda cm\right)$$
$$+\ \ldots\ =0$$

$$\lambda_1 = \frac{g}{2l},\quad \lambda_2 = \frac{2g}{l},\quad \lambda_3 = \frac{(11-\sqrt{57})}{4}\frac{g}{l},\quad \lambda_4 = \frac{11+\sqrt{57}}{4}\cdot\frac{g}{l}$$

$$\lambda_1 = \frac{g}{2l} = \frac{c}{m}\quad \begin{pmatrix} 1{,}5c & c & -1/4\,mg & 0 \\ c & 1{,}5c & 0 & -1/4\,mg \\ -1/4\,mg & 0 & 1/4\,mgl & 0 \\ 0 & -1/4\,mg & 0 & 1/4\,mgl \end{pmatrix} u_1 = 0$$

$$\begin{pmatrix} 1{,}5\frac{mg}{2} & mg/2 & -1/4\,mg & 0 \\ mg/2 & 1{,}5\,mg/2 & 0 & -1/4\,mg \\ -1/4\,mg & 0 & 1/4\,mgl & 0 \\ 0 & -1/4\,mg & 0 & 1/4\,mgl \end{pmatrix} \to \begin{pmatrix} \frac{3}{4}mg & \frac{1}{2}mg & -\frac{1}{4}mg & 0 \\ mg/2 & 1{,}5mg & 0 & -\frac{1}{4}mg \\ 0 & 1{,}5mg & 1/4\,mgl & 1/4\,mgl \\ 0 & -1/4\,mg & 0 & \end{pmatrix}$$

$$\to \begin{pmatrix} 3/4\,mg & \frac{1}{2}mg & -1/4\,mg & 0 \\ 0 & 7/4\,mg & 0 & -\frac{3}{8}mg \\ 0 & 1{,}5mg & 1/4\,mgl & -\frac{1}{4}mg \\ 0 & 0 & 0 & 1/4\,mgl \end{pmatrix} \;\ldots$$